NAME: **MANOJ KUMAR**



Designation: Sr. Technical Lead
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Experience: 8 years
Location: Noida, India

PROFESSIONAL SUMMARY

- ☐ With 8 years of experience in Data Science and Generative AI, and a total of 10+ years of overall experience, I have successfully implemented full -stack NLP, and Generative AI projects using the MLOps Pipeline to make data-driven decisions. I excel in Neural Networks, Transformers, Python, Lang Chain, Lang Graph, RAG, Vector Databases, Hugging Face, and building reliable LLM applications.
- ☐ 0.8 years of experience with Coforge.
- ☐ Experience in RestAPIs: **FastAPI, Flask**.
- ☐ Expertise in using **Gen AI, NLP, ML** and framework like **langchain, langgraph, langsmith, Vector Database, pytorch, tensorflow, scikit-learn.**
- ☐ Experience in working with vector database **Pinecone, FAISS, Chroma DB, Qdrant**.
- ☐ Strong Experience with **Object Oriented Programming Concepts (OOPS).**
- ☐ Working experience of **MLOps** with **GIT, Docker, Kubernetes, DVC, MLflow, Daghub, GitHub Actions.**
- ☐ Experience in RDBMS databases like **SQL, Mysql.**
- ☐ Experience with IDE tools **VS Code, Pycharm, visual Studio.**

TECHNICAL SUMMARY

Technologies/ Frameworks	Python, Machine Learning, Deep Learning: (ANN | Pytorch), Natural Language Processing: (RNN | LSTM | Encoder Decroder | Attention |Transfromer | BERT | GPT), Generative AI: (Open AI, Llama , Gemini, Embeddings, Hugging Face, RAG, Finetuning, AI Agents), LLM Framework: (Langchain | Lang Graph | Llama Index | Lang Smith), Vector Databases: (FAISS | Chroma DB | Pinecone), RestAPIs: (FastAPI | Flask), Databases: (MySQL | SQL), MLOps: (GIT | Docker | Kubernetes | DVC | MLflow | Kubeflow | GitHub Actions), Cloud: (AWS | Azure)
Domain/Industry	Artificial Intelligence for IT Service Management (AI for ITSM), E-Learning, Energy Trading / Commodity Market Analytics ,IT, Finance, HR, Insurance

Build Tools	Python, C#
Internet Technologies	HTML, Jquery, Java script
Web/Enterprise Servers	Microsoft IIS, Nginx
Databases	Sql, mysql
Version Control	DVC, MLflow, Git Hub
Loggers	Python logging module, TensorBoard, MLflow, Prometheus + Grafana, langsmith
IDE	VS Code, Pycharm, Visual Studio, JupyterLab / Jupyter Notebook, Google Colab, Kaggle Kernels
Operating Systems	Windows, MAC, Linux
Software Methodologies	CRISP-DM, MLOps, Agile/Scrum

CLIENT / PROJECT DETAILS

Data scientist | *Fiserv via Coforge Limited | Jan 2026 - Present*

- Develop fraud detection models using CatBoost and GNN architectures for the Value-Added Services layer of the Fiserv payment gateway.
- Built solutions for clients including Clover, The Home Depot, Disney, Wawa, Snappy, CardConnect, and CardPointe.
- Designed combined-model inference logic, integrated graph data through Neptune, and optimized production prediction flows.
- Applied PCA for feature reduction, correlation analysis, and improved model efficiency.
- Applied KMode clustering and undersampling to train data efficiently.
- Improved model accuracy by increasing TP by 200% and decreased FP by 66%. Also improved the latency by 40%.
- Created Docker-based deployments and conducted scalable stress testing up to 1,000 concurrent users.
- Designed MVE and MME strategies to support customer onboarding, startup optimization, and cost efficiency.

Client Name : HCL	**Duration :** 06-25 to 08-25
Team Size	1
Technologies	OpenAI, Pinecone, LangChain, FastAPI, LangSmith, Docker, Python, Azure
Role	Gen AI Eng.

Project Overview	Developed an intelligent chatbot powered by Retrieval -Augmented Generation (RAG) to assist users in querying detailed information about a specific enterprise application and its associated support workflows. The chatbot is capable of answering questions related to how support tickets are raised, categorized, escalated, and resolved across different operational teams, thereby improving user self -service and reducing dependency on manual support.
Roles & Responsibilities	• Worked as a Developer, responsible for the end -to-end design and implementation

Client Name : HCL	**Duration :** 04-25 to 06-25
Team Size	1

Technologies	Pythin, OpenAI,Pinecone, LangChain,Lang Smith,Fast API ,Docker, Azure,Github
Role	Gen AI Eng.
Project Overview	Developed an AI-driven chatbot leveraging Retrieval-Augmented Generation (RAG) to retrieve and deliver the most contextually relevant responses from a question–answer dataset. The system employs advanced natural language processing to understand user intent, match queries with precise answers, and generate accurate responses in real time. This solution enhances enterprise knowledge retrieval, automated customer support, and FAQ management, significantly improving query resolution efficiency and accuracy.
Roles & Responsibilities	• Worked as a Developer, responsible for the end-to-end design and implementation.

Client Name : HCL	**Duration :** 01-25 to 04-25
Team Size	2
Technologies	Python, FastAPI, spaCy, BERT
Role	Data scientist
Project Overview	Developed an AI-driven customer support routing system that automatically classifies and assigns customer queries to the correct department (HR, Finance, IT, etc.) using a hybrid NLP pipeline. Integrated Named Entity Recognition (NER) for keyword extraction and a BERT-based intent classifier for contextual understanding. Implemented fusion logic to combine both signals.
Roles & Responsibilities	• End-to-End Solution Ownership: Designed, developed, and deployed a full-scale AI-driven customer support routing system capable of automatically classifying and assigning customer queries to the right department (HR, Finance, IT, etc.). • Hybrid NLP Pipeline: Engineered a hybrid pipeline combining Named Entity Recognition (NER) for entity extraction and a BERT-based intent classifier for contextual understanding.

	• Fusion Logic Implementation: Developed intelligent signal fusion logic to combine NER and intent classifier outputs, enhancing precision and recall for department classification.

Client Name : HCL		**Duration :** 01-25 to 04-25
Team Size	2	
Technologies	Python, BERT-base-uncased, LORA, PyTorch	
Role	Data scientist	
Project Overview	Developed a deep learning–based multiclass text classification model to automatically categorize user-generated IT support messages (e.g., application access issues, attendance marking failures, app malfunctions) into predefined categories such as SSD, LMS, VSD, and HR. This solution assists users by recommending the appropriate category while raising tickets, improving routing accuracy and reducing manual effort.	
Roles & Responsibilities	• Worked as a Developer, responsible for the end-to-end design and implementation.	
Client Name : NTPC		**Duration :** 01-21 to 06-21
Team Size	3	
Technologies	Python, BERT-base-uncased, LORA, PyTorch	
Role	Data scientist	
Project Overview	Developed a predictive pricing model to align coal auction prices with real-time market demand, improving pricing accuracy	

	and decision-making. Applied regression and time series techniques to forecast demand and recommend optimal starting bids. Enabled real-time insights for auction planning, reducing pricing inefficiencies and increasing buyer Particiation. Designed the solution for scalability across mines and states with minimal customization. Delivered actionable intelligence dashboards for stakeholders to monitor demand trends and adjust strategy dynamically.
Roles & Responsibilities	• Worked as a Developer, responsible for dashboard design and model testing activities. • Designed interactive dashboards to v i sualize key performance indicators (KPIs) such as demand forecasts, auction prices, and participation trends for business stakeholders. • Tested and validated predictive models to ensure forecasting accuracy and model robustness under different market scenarios. • Ensured data consistency and performance optimization for real-time monitoring and reporting.

Client Name : NTPC	**Duration :** 07-21 to 09-21
Team Size	2
Technologies	Python, GPT(Open AI/ LLM), Qdrant, Lang Chain, Fast API
Role	Data scientist
Project Overview	Designed and deployed a production-ready AI chatbot for NTPC to streamline customer inquiries related to its core business areas: Coal Mines, Coal Stations, Gas Stations, Renewable Energy, and Hydro Stations. The chatbot provides fast, accurate, and context-aware responses to user queries by integrating data directly from NTPC's public information sources.

Roles & Responsibilities	• Worked as a Developer, responsible for the end-to-end design, development, and deployment of a production-ready.
Client Name : Wayinfotech Software Solutions Pvt. Ltd.	**Duration :** 01-20 to 03-20
Team Size	2
Technologies	Python, PyTorch, Sense2Vec, T5-Base Pretrained Summarizer, Pretrained T5 Question Generator, MSMARCO-DistilBERT-Base-V2, FastAPI
Role	Data scientist
Project Overview	Developed an AI-powered multiple-choice question (MCQ) generation system that automatically creates objective questions — including multiple correct answers — from a given text or chapter. The solution enhances student engagement, promotes critical thinking, and assists educators in designing high-quality assessments efficiently with minimal manual effort
Roles & Responsibilities	• Worked as a Developer, responsible for the end-to-end design and development of the AI-powered MCQ generation system. • Collected, cleaned, and preprocessed textual data to create a structured dataset for training and evaluation. • Developed REST APIs using FastAPI for real-time quiz generation and integration with e-learning platforms. • Containerized and deployed the solution using Docker/Kubernetes for scalability and efficient model serving. • Conducted model evaluation and optimization to ensure high-quality question generation and contextual accuracy.

Client Name : Wayinfotech Software Solutions Pvt. Ltd.	Duration : 10-20 to 11-20
Team Size	2
Technologies	Python, LSTM, Flask
Role	Data scientist
Project Overview	Developed a Sentiment Analysis Model using TensorFlow, LSTM, and FastAPI to classify text as positive or negative. This model processes user-generated text and predicts the emotional tone using deep learning techniques.
Roles & Responsibilities	• Worked as a Developer, responsible for the end -to-end design, development, and deployment of the Sentiment Analysis Model .

EDUCATIONAL QUALIFICATIONS

- ☐ MSC. IT from swami vivekanand subharti university, meerut - 2022.
- ☐ **Intermediate** from Giridih College - 2009.
- ☐ **High School** from Girls' project high School - 2007.

VISA DETAILS

Country	Visa Type	Validity (MM-YYYY)
